Tarun Yadav

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Work Experience

SDE Intern – Synapse Technologies Aug 2025 – Nov 2025

Bhopal

- Contributed to a real-time multiplayer gaming platform, implementing Redis-based caching and Kafka event-streaming that cut average response latency by 30% across concurrent sessions.
- Increased transaction throughput by 40% by building an asynchronous event-processing pipeline for the core game transaction engine, ensuring data consistency under concurrent load.
- Supported backend reliability during peak load by implementing distributed messaging and caching services, helping the platform maintain high availability under high-concurrency gameplay.

Projects

Distributed Web Crawler – Cluster-Wide Adaptive Politeness Enforcement GitHub

Python 3.13 (asyncio, aiohttp), Redis, SQLite, Docker, Prometheus, Hexagonal Architecture

- Replaced coordinator-based domain assignment with consistent hashing (128 virtual nodes per physical node), resolving ownership locally in $\sim 2\mu s$ with no central lookup and cutting domain reshuffling to $\sim$15.5% on membership changes, versus $\sim$83% under naive modulo hashing.
- Replaced fixed-rate crawling with per-domain AIMD control – the same additive-increase/multiplicative-decrease approach TCP uses for congestion control – after a deliberately throttling test server rejected fixed-rate crawling outright (0% acceptance); AIMD brought that to 94% while holding $\sim$1,280 pages/sec across the cluster, covered by 2,143 tests at 93.4% coverage under strict mypy typing.

AlgoLab – Production-Grade Graph Algorithms Library GitHub

Python, mypy (strict), Algorithm Design, Unit Testing

- Architected a Python graph algorithm library (20 algorithms, 10 families) processing 565k+ edges/second on 50k-node graphs – strict mypy across 65 modules, 7,000+ lines, zero type-ignore comments.
- Engineered dual max-flow implementations (Edmonds-Karp, Dinic's) delivering 40x throughput on parallel-path networks, validated by 970 tests in 1.2s with full determinism.
- Implemented Binary Lifting LCA – O(N log N) preprocessing of 65k-node trees in 253ms, O(log N) queries averaging $3.1\mu s$ – plus a Union-Find with path compression at $0.19\mu s$ amortized finds on 1M-element sets.

Barista AI – Multi-Agent Portfolio Risk Management Platform GitHub

FastAPI, Next.js, Python, Google Gemini, NumPy/SciPy, Docker

- Orchestrated 6 specialized agents (risk analysis, market monitoring, rebalancing) sharing a singleton data-fetcher/cache layer, with Google Gemini for natural-language insights.
- Computed portfolio risk via parametric, historical, and Monte Carlo VaR (10,000 simulations), Sharpe/Sortino, and CVaR with NumPy/SciPy, backed by a rate-limited multi-source fallback chain.

Education

B.Tech in Computer Science & Engineering – Vellore Institute of Technology Expected Sep 2026

- CGPA: 7.7/10 | Coursework: Data Structures & Algorithms, DBMS, Computer Networks, System Design, Operating Systems

Technical Skills

Languages:	Python, TypeScript, JavaScript, SQL
Web & APIs:	React.js, Next.js, Node.js, Express.js, FastAPI, REST APIs, WebSockets
Databases:	PostgreSQL, MySQL, Redis, SQLite, Kafka
Cloud & DevOps:	AWS (EC2, S3, Lambda), GCP (Vertex AI), Docker, Kubernetes, CI/CD, Nginx, Git
Core CS:	Data Structures & Algorithms, OOP, System Design, DBMS, Operating Systems, Computer Networks
AI/ML Tooling:	LangChain, RAG Systems, Prompt Engineering, Multi-Agent Systems

Certifications & Programs

McKinsey Forward Program – McKinsey & Company June 2026

Advanced Software Engineering Job Simulation – Forage (Midas) April 2026

Cloud Computing – NPTEL April 2024